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# *International Mathematics Assessments for Schools (2024-2025)*

## Middle Primary Division Round 2

Time: 120 minutes

*Printed Name* _____ *Code* _____ *Score* _____

## Instructions:

- Do not open the contest booklet until you are told to do so.
- Be sure that your name and code are written on the space provided above.
- Round 2 of IMAS is composed of three parts; the total score is 100 marks.
- Questions 1 to 5 are given as a multiple-choice test. Each question has five possible options marked as A, B, C, D and E. Only one of these options is correct. After making your choice, fill in the appropriate letter in the space provided. Each correct answer is worth 4 marks. There is no penalty for an incorrect answer.
- Questions 6 to 13 are a short answer test. Only Arabic numerals are accepted; using other written text will not be honored or credited. Some questions have more than one answer, as such all answers are required to be written down in the space provided to obtain full marks. Each correct answer is worth 5 marks. There is no penalty for incorrect answers.
- Questions 14 and 15 require a detailed solution or process in which 20 marks are to be awarded to a completely written solution. Partial marks may be given to an incomplete presentation. There is no penalty for an incorrect answer.
- Use of electronic computing devices is not allowed.
- Only pencil, blue or black ball-pens may be used to write your solution or answer.
- Diagrams are not drawn to scale. They are intended as aids only.
- After the contest the invigilator will collect the contest paper.

**The following area is to be filled in by the judges;**
**the contestants are not supposed to mark anything here.**

| Question | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | Total Score | Signature |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Score | | | | | | | | | | | | | | | | | |
| Score | | | | | | | | | | | | | | | | | |

# Middle Primary Division Round 2

## Questions 1 to 5, 4 marks each

1. Andy and Ben go on an 8 km trip using one bicycle shared between them. For each of them, their walking speed is 4 km per hour and their cycling speed is 16 km per hour. They both start at the same point and leave at the same time towards the same destination, where Andy rides the bicycle, while Ben walks. At a certain point, Andy dismounts and leaves the bicycle behind, and then walks to the destination. When Ben reaches the bicycle, he mounts it immediately and rides it for the rest of the trip. Knowing that Ben and Andy arrive at the destination at the exact same time, how many minutes does the whole trip take for each of them?

   （A）30　（B）45　（C）75　（D）90　（E）150

   Answer：_____

2. There are eleven boxes, where five are green, five are red and one is blue. Each of these boxes contains exactly 15, 16, 17, 18, 19, 20, 21, 22, 23, 24 and 25 balls, in some order. It is known the total number of balls inside the green boxes is 30 more than the total number of balls inside the red boxes. What is the number of balls inside the blue box?

   （A）16　（B）18　（C）20　（D）22　（E）24

   Answer：_____

3. Renard wants to completely cover both a $49\times51$ and a $99\times101$ grid by using some number of pieces of a same rectangle having integer side lengths. What is the largest possible area of this rectangle such that both grids can be fully covered without overlapping pieces?

   （A）2　（B）3　（C）4　（D）5　（E）6

   Answer：_____

4. What is the sum of all the digits of the product $142857\times2025$?

   （A）42　（B）43　（C）44　（D）45　（E）46

   Answer：_____

5. The sum of the dots on each pair of opposite faces of a fair die is 7. Alice rolls three fair dice and records the numbers of dots on the top faces (from left to right) as a three-digit number. She also records the numbers of dots on the bottom faces (from left to right) as another three-digit number. Then Alice appends the second number after the first one to form a six-digit number. If this six-digit number is a multiple of 3895, what is value of the first three-digit number?

   （A）234　（B）324　（C）342　（D）423　（E）432

   Answer：_____

## Questions 6 to 13, 5 marks each

6. A store sells special Pokémon stickers and each special Pokémon sticker is put into a special pack. This store also recycles these special packs. Anyone who returns 7 special packs can get one new special Pokémon sticker, which is also put inside a special pack. Chris has purchased 225 special Pokémon stickers from this store. What is the maximum number of extra special Pokémon stickers Chris can get from this store by returning special packs?

Answer : _____ stickers

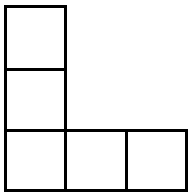
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7. Amy selects a number from 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10. After that, Brandon selects a number from the remaining nine numbers such that the last digit of the product of their numbers is 6. How many ways are there for Amy and Brandon to choose their numbers?

Answer : _____ ways

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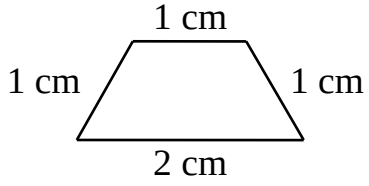
8. In the grid below, we place each of the digits from 1 to 5 in each cell, where each digit is used exactly once. How many different ways are there to place these digits so that the sums in the row and column are identical?



Answer : _____ ways

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9. An equilateral triangle of integral side length, in cm, is said to be "***good***" if we can use some number of pieces of an isosceles trapezoid of side lengths 1 cm, 1 cm, 1 cm and 2 cm, to complete cover the triangle.
(Note that the isosceles trapezoids join side by side and cannot overlap.)

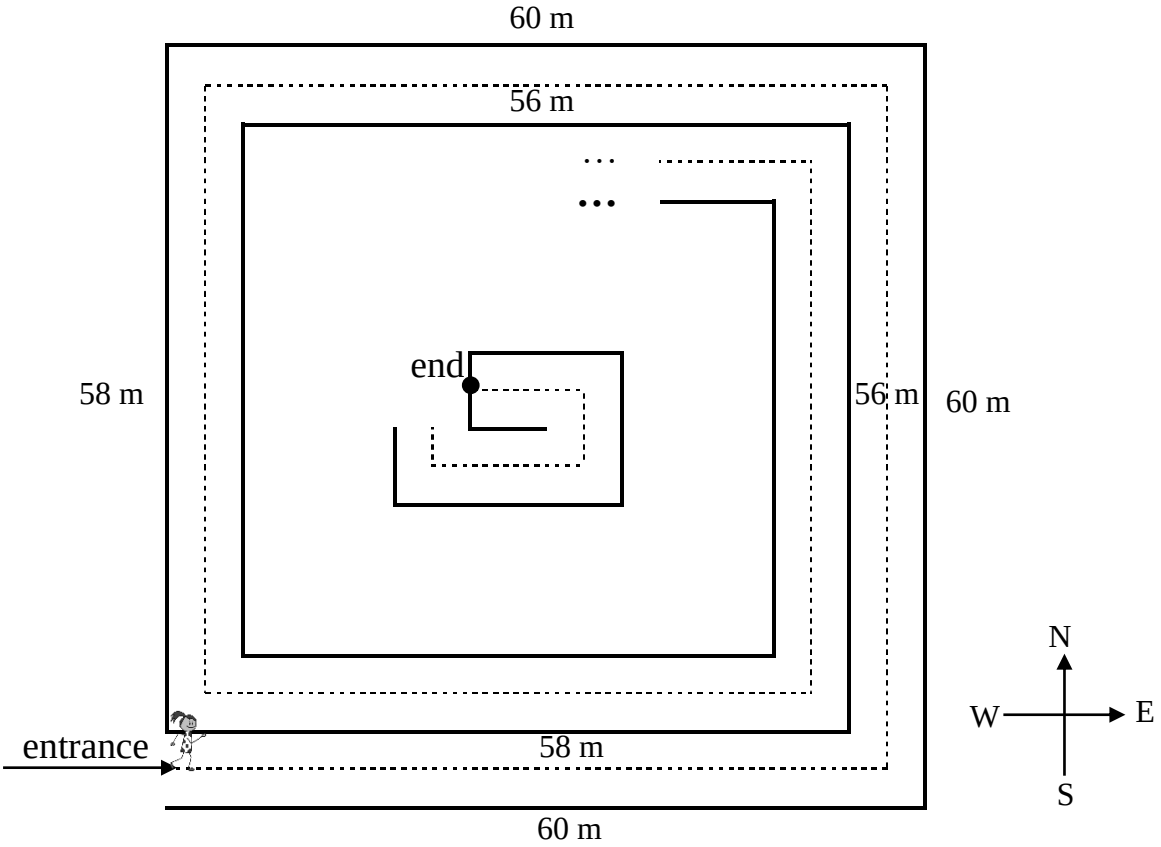



What is the minimum possible side length, in cm, of a "good" triangle with side length equal to or greater than 10 cm?

Answer : _____ cm

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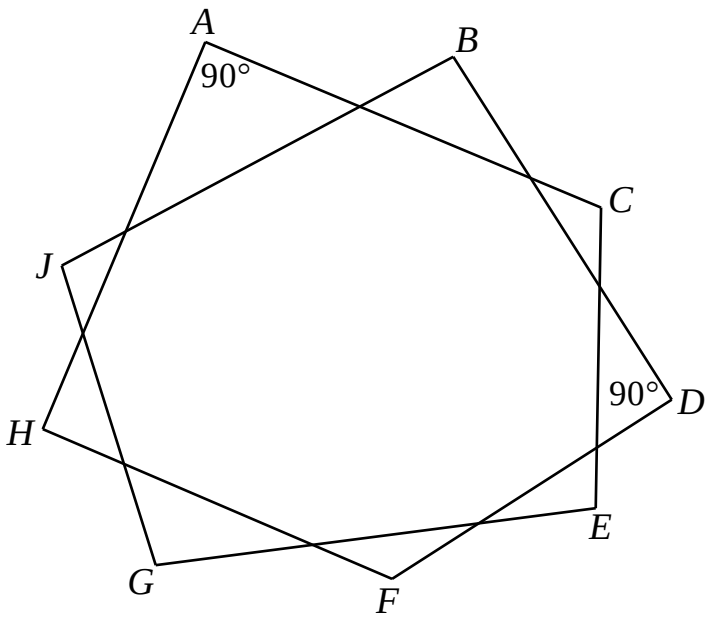
10. A gardener planted rows, which are represented by the solid segments shown in the diagram below, of high bushes in a park. First, he planted a row running 60 m east. Then from the end of that line he planted a row 60 m north, then 60 m west, 58 m south, 58 m east, 56 m north, 56 m west, and so on, where the lengths of the rows are reduced by 2 m after every two anti-clockwise 90-degree turns. This made a square spiral path which is 2 m wide. One day, Betty walked along the middle of the path starting from the entrance to the end, as represented by the thin dashed segments shown in the diagram below. What was Betty's total walking distance, in m?




Answer : _____ m

## MP 4

11. Let *ABCDEFGHJ* be a nine-pointed star, as shown in the diagram below, where $\angle A = \angle D = 90°$.




What is the sum, in degrees, of $\angle B + \angle C + \angle E + \angle F + \angle G + \angle H + \angle J$ ?

Answer： _____ °

12. What is the smallest integer larger than 2025 which starts with 20, ends with 25 in decimal representation and is a multiple of 2025?

Answer： _____

13. A special calculator is capable of only two operations: addition by 1 or multiplication by two. The number 0 was first entered into the calculator, and after some operations, the calculator came up with a result of 2025. What is the smallest number of operations the calculator has to perform in order to obtain this result?

Answer： _____ operations

## Questions 14 to 15, 20 marks each
## (Detailed solutions are needed for these two problems)

14. A box contains 100 cards, where each card is labelled with the integers from 1 to 100. If the cards are drawn one at a time without putting back, what is the minimum number of cards that must be drawn to guarantee that, among the cards selected, the label of one of the cards is exactly twice the label of another card? (If your answer is $n$, write the $n-1$ integers of your choice such that no integer is exactly twice of another integer.)

Answer _____ cards.

If answer is $n$, then none of the following $n-1$ integers is exactly twice of another integer:

_____

15. The integers from 1 to 16 have been placed in a $4\times4$ grid paper, as shown in the diagram below. David cuts this grid paper along the thick grid lines to separate the grid paper into two pieces so that the sum of numbers in each piece is exactly $68=\frac{1+2+3+...+16}{2}$. Find as many other solutions as possible that separates this grid paper into two pieces along the grid lines so that the sum of the numbers in each piece is 68. Draw the cutting line on the diagrams provided.

| | | | |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

Note that if some regions have common points but do not have any common side, then these regions are not in the same piece. For example, the following way of cutting separates the grid paper into three pieces, even if the sums of the numbers in the shaded part and white part are both 68.

| | | | |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

| | | | |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

| | | | |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

| | | | |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

| | | | |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

| | | | |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

| | | | |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

| | | | |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

| | | | |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

| | | | |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

| | | | |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

| | | | |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

| | | | |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

| | | | |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

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